MARTINUS ALVIN HARYANTO

martinus.haryanto@binus.ac.id | +62 882 3900 3390 | Portofolio | Github | Linkedin

EDUCATION

BINUS University *Jakarta, Indonesia*
B.S. Computer Science, Data Science Specialization Track *2024 – 2028 (Expected)*
GPA: 3.96 / 4.00

PUBLICATIONS

"Dynamic Classifier Selection using Local Class Accuracy on Overlap Cleaned Heterogeneous Pool of Experts"
1st Author | Published | Paper and Source Code *Oct 2026 (Expected)*

- Led development of DCS-LCA, a dynamic classifier selection framework combining KNN overlap cleaning with expert selection across Naive Bayes, XGBoost, and SVM, and authored the manuscript; cut log loss by 70.1% (0.898 to 0.269) versus a baseline probability harvesting and reached 91.62% accuracy (macro F1 0.8638) on the datasets.

ORGANIZATIONS

Himpunan Mahasiswa Teknik Informatika (HIMTI) BINUS University
Activist in Publication and Marketing *Oct 2024 – Oct 2025*

- Coordinated end-to-end logistics for three School of Computer Science programs (TECHNO, SESVENT, FEAST 2025), aligning event teams, schedules, and execution requirements.

Freshmen Partner and Freshmen Leader *Sept 2025 – Sept 2026*

- Led orientation and team-building sessions for 40–50 freshmen in the Freshmen Year Program, coordinating facilitators and mentoring them through campus systems and academic culture.

PROJECTS

DataFlow Simulation — Simulated E-Commerce Data Platform with Event-Driven MLOps *Aug 2026*
Dashboard | GitHub

- Built a medallion warehouse (Raw → Silver → Gold) on PostgreSQL with dbt marts for sales, logistics SLA, and user funnels, orchestrated by a 6-DAG Airflow pipeline (load → dbt → Supabase sync → ML train/inference) that gates retraining to every 6th batch.
- Trained Optuna-tuned XGBoost churn and conversion models behind a champion/challenger registry that only promotes models beating both a baseline and the current champion, and shipped a Next.js + Supabase dashboard for pipeline health, model performance, and predictions.

NutriVision Pro — Computer Vision & MLOps Cloud Platform *Sept 2026*
Live Web App | Github

- Deployed a Computer Vision backend on AWS EC2 and S3 with Docker Compose, serving a fine-tuned YOLOv11 instance segmentation model for real-time food scanning and nutrition analysis, with a zero-downtime CI/CD pipeline on GitHub Actions and AWS ECR (image builds, registry sync, secure .env injection).
- Engineered a secure FastAPI service with admin header validation (x-admin-token), Redis caching, and dynamic ONNX model reloading that pulls weights from a private S3 bucket on startup, with fallback verification and health-check endpoints.

YouTube Performance Predictor — Multimodal ML & MLOps Platform *Oct 2026*
Live Web App | Github

- Built a multimodal ML service that scores a YouTube video against its channel's recent average from the thumbnail and title alone, trained on 11,285 videos (135 channels) with chronological splits and subscriber features removed to prevent target leakage.
- Designed a token-level early-fusion model over frozen CLIP and tabular features, tuned with a validation-only Optuna search; a 5-seed ensemble reached 0.431 test Spearman, and a single model beat a tuned late-fusion baseline by +0.099 (0.373 vs. 0.274).
- Deployed it as a FastAPI service on an Oracle Cloud ARM VM (Docker Compose, Caddy, API-key auth) with a Vercel frontend, Airflow ingestion and drift monitoring, and SHA-256-pinned model bundles with single-step rollback.

TECHNICAL SKILLS

Programming: Python, SQL, Java, C, R
Data Engineering: Apache Airflow, dbt, PostgreSQL, pgvector, MinIO, Supabase, Pandas, Medallion Architecture
Machine Learning & Deep Learning: Scikit-Learn, XGBoost, LightGBM, PyTorch, Multimodal Early Fusion (Cross-Attention, Transformers), Computer Vision (YOLOv11, CLIP, Grounding DINO), NLP (BERT), Optuna
MLOps & Cloud: Docker, GitHub Actions, CI/CD, Continuous Training Pipelines, Drift Monitoring, FastAPI, Caddy, AWS, Oracle Cloud, Vercel, Firebase